

Aura Tesla Opt — abstract and full operation prediction

Device: Aura Tesla Opt — DIN-rail home energy hub (27 BOM rows, \$1,072) · **Agent:** Aura (local LLM, resident) · **Status:** hardware design closed at round 5 · **Source:** Blueprint.io export, 33 electrical connections.

Abstract

Aura Tesla Opt is a wall-mounted, DIN-rail energy hub that turns a European household's largest loads into a scheduleable resource. A certified Shelly Pro 3EM meters all three phases through CT clamps; four certified Shelly Pro 1PM switches (16 A each) control the water heater, wall heater, EV charger and an auxiliary circuit; a fanless N100-class mini-PC hosts **Aura**, a resident agent built on a local LLM whose proposals pass a deterministic safety policy before anything is switched. The electronics sit on a 24 V DC bus backed by a 240 Wh LiFePO4 pack — roughly 19 h of host ride-through — with a DC-DC converter feeding the 12 V host and network switch. Aura runs a closed loop: hourly planning ticks against live telemetry and tariff windows, nightly deterministic habit extraction, and a weekly governed fine-tuning cycle that bakes accumulated human feedback into the model behind an evaluation gate with one-command rollback. The model proposes, policy disposes, the outbox executes — and every decision lands in a journal the owner can read. Predicted outcome: no new behaviour required from the household; 150–400 €/yr of scenario savings from load shifting in year one, rising as habit profiles and adapter quality mature.

Apstrakt (srpski — latinica)

Aura Tesla Opt je zidni, na DIN šinu montiran energetska hub koji pretvara najveće kućne potrošače u resurs kojim se može upravljati po rasporedu. Sertifikovani Shelly Pro 3EM meri sve tri faze preko CT klemu; četiri sertifikovana Shelly Pro 1PM prekidača (16 A) kontrolišu bojler, zidni grejač, punjač električnog vozila i pomoćno kolo; fanless mini-PC klase N100 hostuje **Auru**, rezidentnog agenta zasnovanog na lokalnom LLM-u, čije predloge propušta deterministički sigurnosni sloj pre nego što se bilo šta prekine ili uključi. Elektronika radi na 24 V DC sabirnici podržanoj LiFePO4 baterijom od 240 Wh — otprilike 19 h autonomije hosta — dok DC-DC konverter napaja host sa 12 V i mrežni svič. Aura vrti zatvorenu petlju: svakog sata planira prema telemetriji i tarifnim prozorima, svake noći deterministički izvlači navike, a svake nedelje vođenim ciklusom fino podešava model, upisujući akumuliranu povratnu informaciju ukućana u težine modela — iza evaluacione kapije sa vraćanjem na prethodnu verziju jednom komandom. Model predlaže, politika odlučuje, outbox

izvršava — a svaka odluka završava u dnevniku koji vlasnik može da čita. Predviđeni ishod: od ukućana se ne traži nikakva promena ponašanja; 150–400 €/god uštede pomeranjem potrošnje u prvoj godini, sa rastom kako profili navika i adapteri sazrevaju.

1. Key figures at a glance

Metric	Value	Source
Bill of materials	27 line items · \$1,072	PARTS.csv
Electrical / mechanical split	\$904 / \$168 (84% / 16%)	PARTS.csv
Electrical connections	33 (27 power, 6 data)	ELECTRICAL_CONNECTIONS.json
Voltage classes wired	230 V AC · 24 V DC · 12 V DC · LAN	ELECTRICAL_CONNECTIONS.json
Controlled loads	4 × Shelly Pro 1PM, 16 A each	PARTS.csv
Battery ride-through	240 Wh LiFePO4 · ≈19 h host	design (24 V bus)
Modelled saving	150–400 €/yr (year one)	MODELLED scenario

2. System architecture

Figure 1 · Aura Tesla Opt — three-plane architecture

SCHEMATIC — topology read from the Blueprint.io export (33 electrical connections).

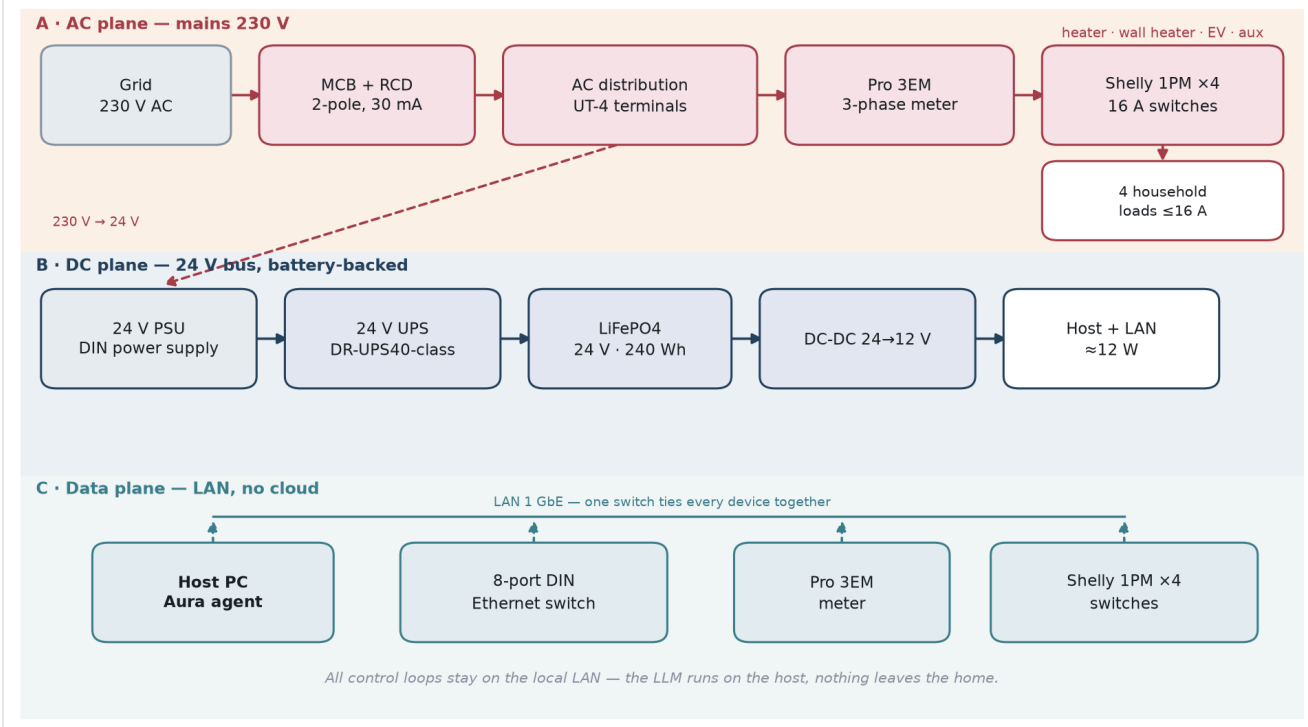


Figure 1. Three-plane architecture — AC (230 V), DC (24 V bus, battery-backed) and data (local LAN, no cloud). *Schematic, topology read from the Blueprint.io export (33 electrical connections).*

3. Power & data distribution

Figure 2 · Power & data distribution — from the export

Measured directly from ELECTRICAL_CONNECTIONS.json (33 connections).

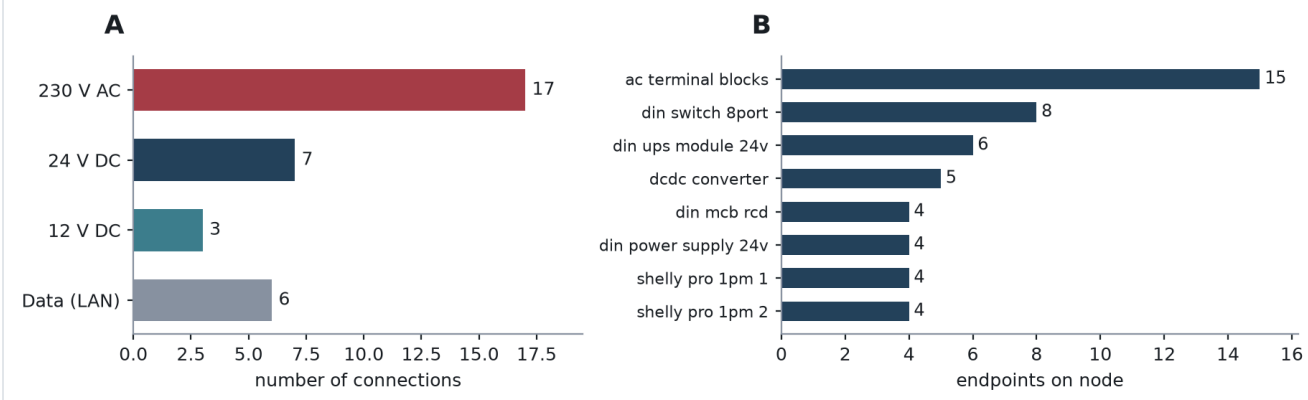


Figure 2. Panel A — connections per voltage class (230 V AC 17, 24 V DC 7, 12 V DC 3, data 6). Panel B — endpoints per node. Measured directly from ELECTRICAL_CONNECTIONS.json (33 connections).

4. Bill of materials

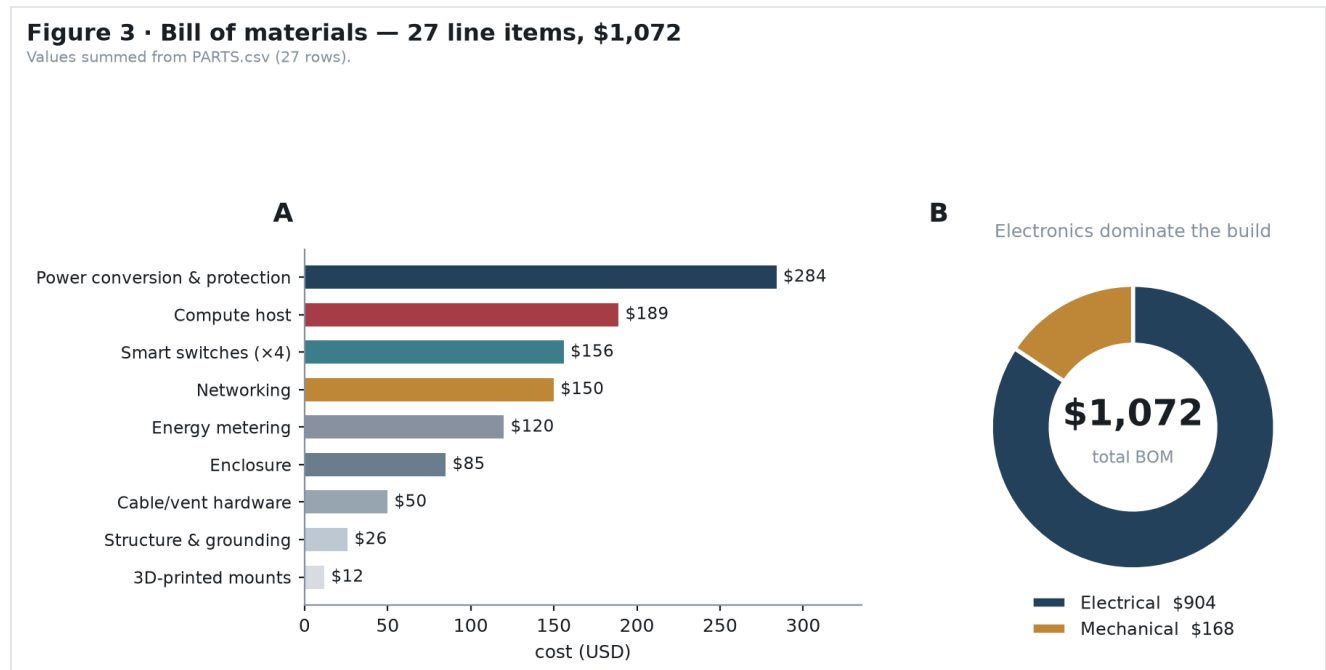


Figure 3. Panel A — cost by part type (power conversion & protection dominates at \$284). Panel B — electrical \$904 vs mechanical \$168. Values summed from PARTS.csv (27 rows).

5. Load-shift scenario

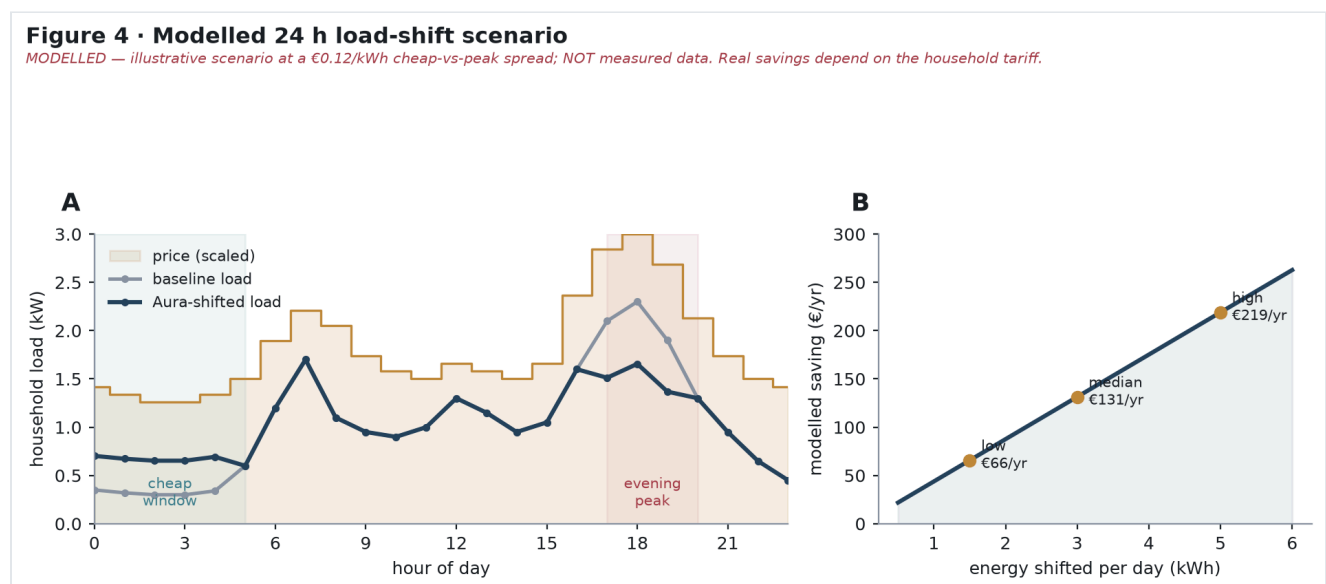


Figure 4. Panel A — modelled 24 h load before and after Aura shifts flexible loads into the cheap night window. Panel B — modelled annual saving versus energy shifted per day. *Modelled* — illustrative at a €0.12/kWh cheap-vs-peak spread; not measured data.

6. Method & reproducibility

Every figure is generated by `make_figures.py` directly from the Blueprint.io export, so each value is traceable to a source file:

- **PARTS.csv** — 27 line items, \$1,072; electrical \$904 / mechanical \$168.
- **ELECTRICAL_CONNECTIONS.json** — 33 connections (230 V AC 17, 24 V DC 7, 12 V DC 3, data 6).
- **MECHANICAL_CONNECTIONS.json** — 26 mounting relationships.

Schematic panels are labelled *schematic*, modelled panels *modelled*; no measured data is claimed where none exists. Companion SVGs (`out/*.svg`) remain editable for later revision.